

Rival model family: common operating point and channels, different dynamic laws

ODE

Caputo

DDE (retarded)

latent m

common experimental reference: coexistence operating point · input/observation channels · physical units

memory law:

$$\dot{\xi} = J\xi + Bu$$

memory law:

$$\tau_0^{\alpha-1} {}^C D_t^\alpha \xi = J\xi + Bu$$

memory law:

$$\dot{\xi} = A_0 \xi(t) + A_1 \xi(t - \tau) + Bu$$

memory law:

$$\dot{q} = \bar{A}q + \bar{B}u, \quad y = \bar{C}q$$

Markov:
no memory

power-law kernel
 $t^{\alpha-1}/\Gamma(\alpha)$

single discrete
delay τ

m exponential
relaxation modes

complexity budget:
 $m \leq m_{\max}$ (declared, not biological)

ODE/Caputo share the ecological Jacobian J ; DDE and latent rivals are calibrated alternative response laws, not identical Jacobian copies.